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Recurrent Memory Transformer

Motivation. Standard Transformers mix local token representations with global context and scale quadratically in sequence length, while Transformer-XL extends context via cached hidden states but entangles token and memory representations and restricts gradient flow across segments.

Core Idea. The Recurrent Memory Transformer (RMT) augments a vanilla Transformer with dedicated memory tokens that are prepended for reading and appended for writing, allowing explicit storage and recurrent propagation of compressed sequence information across segments.

Memory Mechanism. At each segment, memory tokens are processed jointly with input tokens through all Transformer layers, and the output memory tokens are passed forward as the memory state for the next segment, enabling segment-level recurrence without modifying internal Transformer layers.

Architectural Distinction. Unlike Transformer-XL, which caches layer-wise hidden states, RMT uses a fixed number of explicit memory vectors, reducing representation mixing and enabling deeper memory processing as the same memory tokens traverse the network repeatedly.

Backpropagation Through Time. RMT allows gradients to flow across segments via truncated backpropagation through time, improving long-range credit assignment compared to stop-gradient mechanisms in Transformer-XL.

Empirical Findings. On synthetic sequence tasks such as Copy and Reverse, RMT maintains accuracy across long segment chains where Transformer-XL degrades, and on WikiText-103 it achieves competitive perplexity with significantly smaller memory sizes.

Interpretation. RMT can be viewed as a recurrent compression mechanism that separates transient token processing from persistent storage, enabling efficient long-context modeling while preserving compatibility with pretrained Transformer architectures.

Key Takeaway. By introducing explicit, recurrent memory tokens without altering Transformer internals, RMT provides a simple yet powerful framework for long-sequence modeling based on structured memory rather than hidden-state caching.